Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**CS 535**
Program:	**MS(Data Science)**	**Semester:**	**Fall 2021**
Duration:	**30 Minutes**	**Total Marks:**	**20**
Paper Date:	**17-Dec-21**	**Weight**	
Section:	**CS**	**Page(s):**	**3**
Exam:	**Quiz 3**		

BERT Paper: Jacob Devlin,Ming-Wei Chang,Kenton Lee,Kristina Toutanova, **BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding**, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*

Q1) How did the authors of BERT paper preprocess input text for language modeling training? [5 Marks]

Q2) How did the authors of BERT paper handle out of vocabulary words? [3 Marks]

Transformer Paper: Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. **Attention is all you need**. In Proceedings of *the 31st International Conference on Neural Information Processing Systems* (NIPS'17)

Q3) Tranformer model is not based on a sequence based model like RNN or LSTM. It is based on feed forward neural network. Does it mean that it is using bag of words model where order of words does not matter?
Sentence 1 = X A B C Y
Sentence 2 = A B C X Y
The internal vector representation for word Y would be same in both sentences? [5 Marks]

Name: _____ Reg #: _____ Section: _____

Q4) What will be vector representation (output of first attention layer for one attention head) of the word "machines" in following text using the attention mechanism of Transformer paper. [5 Marks]

"Thinking machines like humans"

Query, key and value vectors of all words are given as follows:

	Thinking	machines	like	humans
query	2 3 1 4	3 2 5 1	1 1 3 2	2 7 1 3
key	3 2 5 5	2 1 2 3	6 3 4 2	1 3 8 5
value	2 1 5 3	4 1 3 4	5 4 6 1	2 6 2 1